

Lesson: Lighting

Lighting is one of the most important aspects to film making. We need it to not only record images onto film but to project it as well. We use it as part of our visual storytelling tool set since it can be used to create mood, focus attention, and add mystery to a scene. When drawing storyboard frames, the kind of shadows you create are going to depend on the quality of light. They will also help determine which drawing tools are best for the job. For instance, if you're going to create shadows that have hard edges, you'll need a sharp drawing tool like a well sharpened pencil or chisel-tip marker. If you're working in Photoshop then using a selection tool like the lasso or marquee tool will be best.

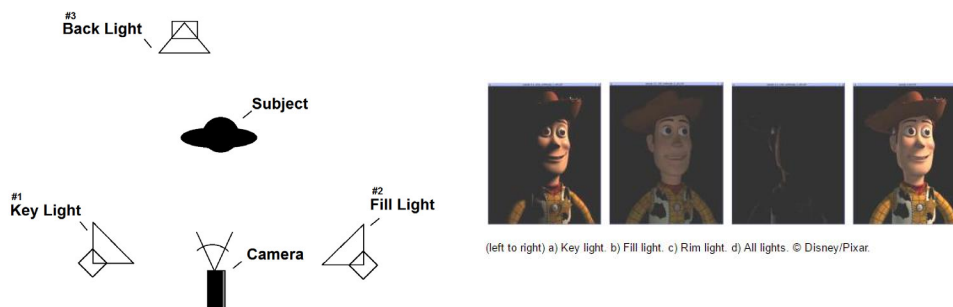
3 Point Lighting

The most basic lighting setup is called Three Point Lighting. It consists of three lights, the key light, the fill light, and the back light.

The **key light** is the main light source and is typically set to 100%. The purpose of the key light is to highlight the form and dimension of the subject. It can be set anywhere in front of the subject along a forty-five degree arc.

The **fill light** is set opposite the key light and is usually between 50-75% of the brightness of the key light. Its purpose is to fill the dark shadows cast by the key light.

The **back light** is placed behind the subject and lights it from the rear. Rather than providing direct lighting (like the key and fill), its purpose is to provide definition and subtle highlights around the subject's outlines. This helps separate the subject from the background and provide a three-dimensional look.



High-key lighting is a style of lighting for film, television, or photography that aims to reduce the lighting ratio present in the scene. This was originally done partly for technological reasons, since

early film and television did not deal well with high contrast ratios, but now is used to suggest an upbeat mood.

High- Key Lighting



Low-key lighting is a style of lighting for photography, film or television. It is a necessary element in creating a chiaroscuro effect. Traditional photographic lighting, three-point lighting uses a key light, a fill light, and a back light for illumination. Low-key lighting often uses only one key light, optionally controlled with a fill light or a simple reflector.

Low Key Lighting



Contrast

High Contrast Lighting emphasizes the the extreme differences between light and dark and is typically seen in thrillers, horror, and suspense films. This type of lighting creates dramatic and interesting images by setting off one part of an image from another.

High Contrast Lighting



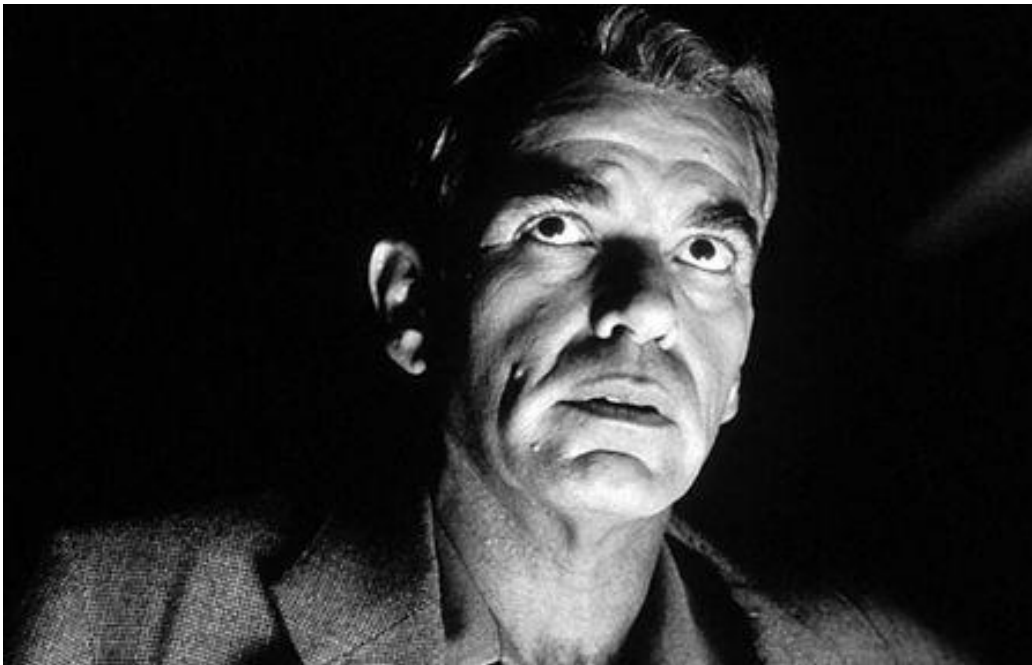
Low Contrast lighting uses a diffused light that creates no pure whites or blacks but rather a wide range of grays. Rather than isolating certain areas it instead allows the viewer to see the overall

composition as a whole but is often considered boring due to its effect of flattening the image.



Lighting Quality

Hard Light is a type of lighting that creates clearly defined shadows and is often used to emphasize the dramatic by illuminating the character's imperfections.



Soft Light creates soft, diffused shadows and is often used to hide any imperfections.



Direction of Light

Frontal Lighting is lighting that is set directly in front of the subject. It creates a flat image with not much distinction between the foreground and background.



Side Lighting lights the subject from the side or sides and creates a strong contrast between light and shadow. This type of lighting is often used to convey the protagonists struggle by being half in light and half in shadow.



Back Lighting is when the light source is directly behind the subject, creating a silhouette.



Under Lighting comes from below and produces odd looking shadows. It is often used to create the appearance of being malevolent or sinister and is often seen in horror films and thrillers.



The Color of Light

The color of light plays an important role in the visual style of film, animation, and game design. Color is used to create a sense of atmosphere or establish a mood, providing an emotional impact. It can tell the viewer how to feel at certain times or direct a player along a path. It can also be used to

indicate an upcoming event. For instance, in the game *The Last of Us* the color yellow is used to direct players where to go and in the film *The Matrix*, green is used to indicate when the characters are plugged into the matrix. Color can also be used to indicate a mood or feeling by adjusting the spectrum from bright, brilliant colors like you would see in upbeat movies, to those that are dark and desaturated such as in horror or suspense films.

Meanings of Color

Black - Authority, power, evil, pessimism.

Red - Optimism, sex, danger, aggression, energy, excitement, passion, fire, love

Green - Envy, fertility, growth, money, luck, healing, success

Pink - tranquility, love, flirtatiousness, softness, delicacy, sweetness

Yellow - cheerfulness, joy, optimism, happiness, energy

Orange - joy, creativity, energy, success, encouragement

Purple - royalty, power, nobility, luxury, spirituality, wisdom, mystery, wealth

Blue - calmness, truth, peace, harmony, confidence, wisdom, loyalty, healing, water, stability

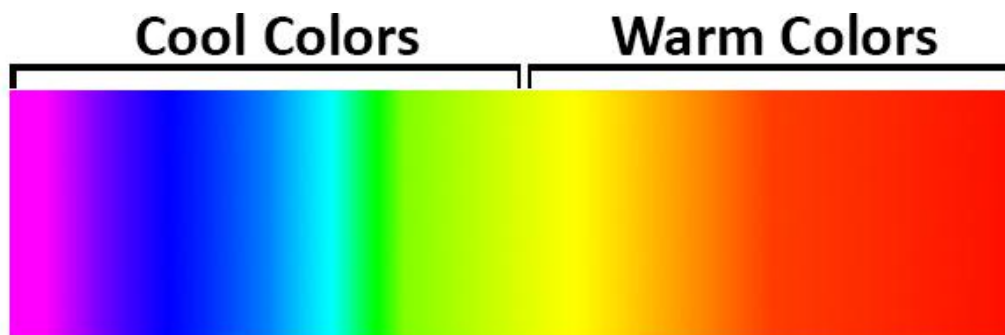
White - innocence, sterility, purity

Gray - neutrality, corporate, quality, practicality

Brown - stability, endurance, simplicity, friendship

Color Temperatures

The temperature of color refers to its indication of warmness versus coolness. Each color on the color spectrum falls within one of these two ranges. Warm colors include red, yellow, and orange because they remind us of warm or hot things. Cool colors include green, blue, and purple and remind us of things that are cool or cold. Fire is typically red while ice is usually blue. These temperatures can be used to convey a certain meaning or mood within the scope of a scene.



Warm



Cool

